

第二届“认证杯”数学中国

数学建模国际赛

承 诺 书

我们仔细阅读了第二届“认证杯”数学中国数学建模国际赛的竞赛规则。

我们完全明白，在竞赛开始后参赛队员不能以任何方式（包括电话、电子邮件、网上咨询等）与队外的任何人（包括指导教师）研究、讨论与赛题有关的问题。

我们知道，抄袭别人的成果是违反竞赛规则的，如果引用别人的成果或其他公开的资料（包括网上查到的资料），必须按照规定的参考文献的表述方式在正文引用处和参考文献中明确列出。

我们郑重承诺，严格遵守竞赛规则，以保证竞赛的公正、公平性。如有违反竞赛规则的行为，我们将受到严肃处理。

我们允许数学中国网站(www.madio.net)公布论文，以供网友之间学习交流，数学中国网站以非商业目的的论文交流不需要提前取得我们的同意。

我们的参赛队号为：1264

我们选择的题目是： B

参赛队员 (签名)：

队员 1：田轲

队员 2：尚笑

队员 3：范庆辉

参赛队教练员 (签名)： 无

第二届“认证杯”数学中国

数学建模国际赛

编 号 专 用 页

参赛队伍的参赛队号：（请各个参赛队提前填写好）：

1264

竞赛统一编号（由竞赛组委会送至评委团前编号）：

竞赛评阅编号（由竞赛评委团评阅前进行编号）：

Dynamic HR Management

Abstract:

Hermit crabs' behavior and the theory of vacancy chains give us a beneficial method which can be applied in the management of HR. This method can improve the effectiveness and control expenditures within a reasonable range at the same time.

In our models, we divide the problem into the static human resources programming, the shortest queue of job applicants and the dynamic model of human resources management. HR allocation of fixed number of employees and jobs is the basis of dynamic management. Firstly, in model I, we define the system of measuring employees' working effectiveness. Based on this, we set up the programming model to achieve the optimal total effectiveness. Secondly, we are inspired by the crabs' behavior of waiting for a certain period of time. Therefore, we establish model II to determine the shortest queue of job applicants. The function of cost-effectiveness is the basis of this model.

Model III is a dynamic model. Initial job vacancies at the beginning of period t are the driving force of HR adjustment. We define the mechanism of this driving force to guarantee the optimal HR allocation. We have referred to the BP neural network algorithm and genetic algorithm. According to the relationship of different periods and the continuous process of optimization of HR, we simulate the process and predict the trend in the future with computers.

In the last, we suggest some possible ways to improve our model, including the difficulty in collecting information about applicants and the change of working effectiveness with time, etc. These are also our direction in the future.

Key words:

Effectiveness optimization, mechanism of dynamic control, random system

Contents

1 Introduction.....	6
1.1 Problem Review.....	6
1.2 Preliminary Analysis.....	6
1.3 Assumptions.....	9
1.4 Variables and Definition	9
2 Model I: Static Human Resources Programming.....	10
2.1 Definition of Indexes	10
2.1.1 Indexes of Individual Working Ability	10
2.1.2 Indexes of Individual Working Preference	11
2.1.3 Indexes of Individual Working Effectiveness	12
2.2 The Primary Model.....	12
2.2.1 The Objective Function.....	12
2.2.2 The Constraint Condition.....	13
2.2.3 The Primary Model	13
2.3 The Intermediate Model.....	14
2.4 The Advanced Model	14
3 Model II: The Shortest Queue of Job Applicants	15
3.1 The Distribution of Human Resources	16
3.2 Effectiveness-cost Model.....	17
3.2.1 Total Effectiveness	17
3.2.2 The Cost Curve	18
3.2.3 The Optimal Length of Applicant Queue.....	18
3.2.4 The Model of Selections of more than one applicants.....	19
4 Model III: The Dynamic Model.....	20
4.1 The Times of Job Exchange	21
4.2 The Shortest Transfer Path of Jobs	22
4.3 The Prediction of Transfer Effectiveness.....	23
4.3.1 Transfer of a Single Employee.....	23
4.3.2 Transfer of more than one Employees	24
4.3.3 Uncertain Transfer of Employees	25
4.4 Predictions of the Change of HR Expenditures	25
4.4.1 The goal of Predictions	25
4.4.2 The Prediction of Transfer Path	25
5 Analysis of Solutions and Result.....	26
5.1 Analysis of the result.....	28
5.2 Solutions of the Problem.....	29
5.2.1 Answer to Question 1.....	29
5.2.2 Answer to Question 2.....	30
6 Evaluation of the Model and Possible Improvements	30
6.1 Advantages of Our Models	30
6.2 Disadvantages of Our Models.....	31
6.3 Possible Improvements	31

6.3.1 Application of Random Arrays	31
6.3.2 The Change of effectiveness index	31
7 Advertising Sheet	33
8 References	34
9 Appendix.....	36

1 Introduction

1.1 Problem Review

When a lone crab encountered one of the beautiful new shells, it behaved specially. After a series of behavior, the resources allocation within the group reaches optimization.

Sociologists and economists use the term “vacancy chain” to describe a sequential exchange of resources that benefits every individual in the sequence. vacancy chains are an excellent way to distribute resources. So it makes sense that hermit crabs have evolved sophisticated social behaviors to make the most of vacancy chains.

Therefore, the goal here is:

1. Evolve social strategies to exchange jobs in queue so that everyone benefits from it. Can we apply them to all industries? If not, what conditions need to meet?
2. Now a new Human Resource website wants to adopt this idea, How many job applicants need to satisfy both employers and employees. Run it to predict the change of HR expends for some enterprise customers within the next five years.
3. Prepare an advertising sheet for the website highlighting the strategies and results.

1.2 Preliminary Analysis

Hermit crabs' behavior of exchange of shelters has significant meanings to the management and optimization of human resources. This behavior is based on the satisfactory of individual demand and tries to distribute resources in the whole society in the most beneficial way, though a dynamic process of adjustment.

Given in the problems, we are required to set up a dynamic model to describe the movement of human resources in a business based on the analysis of hermit crabs' behavior and vacancy chain. There are many reasons causing job vacancies in a

business. Generally, these vacancies will firstly be taken up by existing employees though intra-company transfer of jobs. Then the vacancies left will be filled by new staffs though recruitment from the outside of the business. And then this process will be repeated in the next job adjustment. This process is shown in Figure 1. To sum up, the dynamic movement of human resources can be divided into three steps:

- Step 1: Job vacancies arise. We need to reallocate existing employees within the business to the different jobs to reach optimization.
- Step 2: After the reallocation, the job vacancies left will be taken up by new employees recruited from the society.
- Step 3: when there are new job vacancies and in each business period, repeat stage 1 and 2 above to adjust human resources allocation.

Specifically, step 1 is the issue of allocating human resources under the condition that the number of jobs and employees are constant. We will treat the maximization of the total effectiveness function as our goal. In this essay, we describe this process as the static model of human resources allocation.

Step 2 is mainly used in an open process of recruitment. This includes the determination of the number of job applicants in order to achieve the balance between the best choice of employees and the smallest number of applicants required. The problem of the balance is due to the randomization of probability of each applicant's appearance.

Step 3 is a dynamic model of human resources management and allocation. We have referred to the BP neural network algorithm and genetic algorithm. We set up a dynamic model with multiple input and single output. The flow chart is shown as follows:

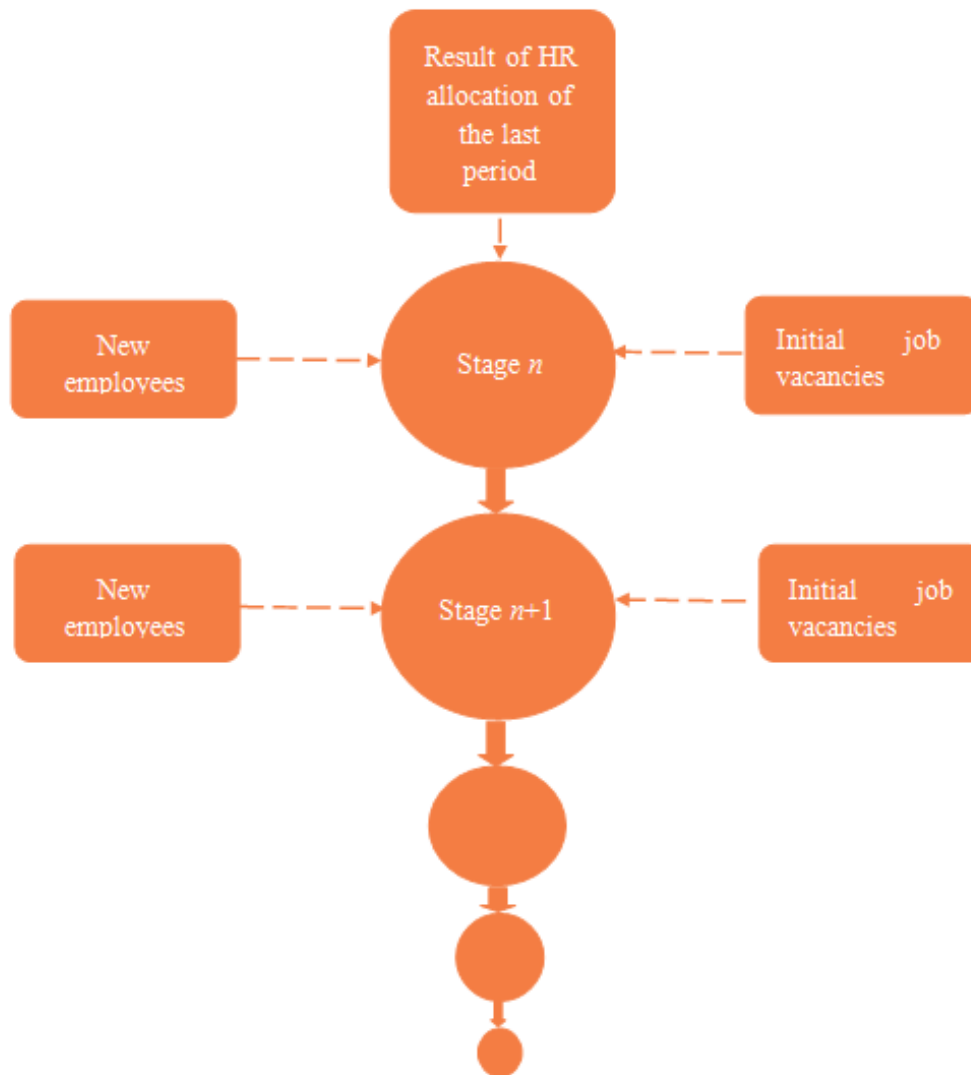


Figure 1: Flow Chart of the Dynamic Model

In this process, we need to consider the following factors:

- The initial job vacancies are the driving force of this dynamic system. The probabilities of job vacancies vary in different business. (Here we define job vacancy as that the previous employee left the job and hence a vacancy arises.)
- The choice of newly recruited employees is based on the demand of the business. Therefore, the individual effectiveness of the newly recruited employees depends on the nature of the job vacancies of the business.
- Upon the appearance of the job vacancies, we should firstly consider to fill these vacancies with existing employees. This means that there will be job exchanges

and reallocation of staffs within the business. This process is similar with the behavior of hermit crabs. Practical situation also approves the process. Most job vacancies arise as a result of retirement and other common factors. These vacancies are usually of high levels and cannot be taken up by new employees. Only when the previous employees get promoted and leave the previous job of low levels vacant, the vacancies can be filled by newly employed staffs. Then in the next period, new employees in this period can have the chance to exchange jobs with existing employees.

1.3 Assumptions

- Expenditures incurred in the process of recruitment is only related to the number of applicants.
- There is no contrived factors and other limitations in the internal adjustment of human resources.
- During the process of recruitment, the employers can know all the information about the applicants.
- There are sufficient applicants in the society.

1.4 Variables and Definition

E	The effectiveness matrix of employees
$E(K)$	the expectation of total times of transfer
$E(Y)$	the expectation of total effectiveness
r	the efficiency of allocation
L	The length of Queue

2 Model I: Static Human Resources Programming

For a group of employees of fixed number and a set of fixed jobs, we need to take account of their ability, their preference to different jobs, the value the employees can bring into the jobs and the extent to which they can influence the performance and efficiency of the group as a whole.

2.1 Definition of Indexes

Before the exchange and reallocation of jobs, the most important task is to learn about employees' ability and preference and try to avoid information asymmetry.

2.1.1 Indexes of Individual Working Ability

As the leaders of a business, it is necessary to make sure that each employee is positioned in the job which he is best at. In terms of the conditions above, we make the following assumptions:

- We can know each employee's working ability on each job through interviews and regular check, etc. And all the information we get are accurate.
- The factor of learning curve effect which can improve employees' ability should be ignored.
- We do not consider the transfer of business and tasks among different jobs.

We divide the level of each employee's working ability into 5 degrees as follows:

Table 1: Level of Working Ability

Degree	A	B	C	D
Score	5	4	3	2

In order to transfer the degree of working ability into available numerical information for analysis purposes, we must know that the level of working ability in practice is nonlinear. According to fuzzy evaluation methods, we apply Cauchy distribute function

to the degree of working ability to do the nonlinear change. The equations are as follows:

$$a = \begin{cases} [1 + A(x - B)^{-2}]^{-1} & 1 \leq x \leq 3 \\ m \ln x + n & 3 \leq x \leq 5 \end{cases}$$

Where A, B, m, n are parameters, x is the value of working ability, a is the value after quantification.

Now we define the following equations:

- When the degree is A and the valuation of working ability is 5, the degree of membership is 1, which means $f(5) = 1$.
- When the degree is C and the valuation of working ability is 3, the degree of membership is 0.8, which means $f(3) = 0.8$.
- When the valuation of working ability is 1, the degree of membership is 0.01, which means $f(1) = 0.01$.

Then we can solve that

$$\begin{cases} A = 1.1086 \\ B = 0.8942 \\ m = 0.3915 \\ n = 0.3699 \end{cases}$$

Apply the value of these parameters into the equation above and we can get that the quantization range of employees' working ability is $\{1, 0.9126, 0.8, 0.5245\}$.

We set up the matrix of indexes of existing employees' working ability as follows:

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1m} \\ a_{21} & a_{22} & \cdots & a_{2m} \\ \vdots & \vdots & \cdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nm} \end{bmatrix}$$

where a_{ij} is the working ability of employee i in job j .

2.1.2 Indexes of Individual Working Preference

Every employee's preference towards work and jobs and their emotion will influence

their performance in the jobs and their effectiveness.

Following the method used in 2.1.1, we set up the matrix of indexes of employees' working preference as follows

$$P = \begin{bmatrix} p_{11} & p_{12} & \cdots & p_{1m} \\ p_{21} & p_{22} & \cdots & p_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ p_{n1} & p_{n2} & \cdots & p_{nm} \end{bmatrix}$$

where p_{ij} is the working preference of employee i in job j .

2.1.3 Indexes of Individual Working Effectiveness

The employees' working effectiveness is valued by the combination of their working ability and preference. We define matrix E as the effectiveness of employees as follows:

$$E = \begin{bmatrix} e_{11} & e_{12} & \cdots & e_{1m} \\ e_{21} & e_{22} & \cdots & e_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ e_{n1} & e_{n2} & \cdots & e_{nm} \end{bmatrix}$$

where e_{ij} is the working effectiveness of employee i in job j .

$$e_{ij} = \alpha \times a_{ij} + (1 - \alpha) \times p_{ij} \quad 0 < \alpha < 1$$

where α is the weight of working ability.

2.2 The Primary Model

In the static programming of human resources, we seek the maximization of group effectiveness and optimization. Base on the analysis above, we take into consideration of employees' working ability and preference towards jobs in the objective function, which is related to the actual conditions. Here we assume that jobs in the business and employees are one-to-one correspondent.

2.2.1 The Objective Function

We define the objective function as follows:

$$X = \begin{bmatrix} x_{11} & \cdots & x_{1n} \\ \vdots & \ddots & \vdots \\ x_{n1} & \cdots & x_{nn} \end{bmatrix}$$

where x_{ij} is a 0-1 variable and its value is as follows:

$$x_{ij} = \begin{cases} 0 & \text{employee } i \text{ does not take up job } j \\ 1 & \text{employee } i \text{ takes up job } j \end{cases}$$

Known from the question, our equation of optimization of effectiveness is

$$\text{MAX} \sum_{i=1}^n e_{ij} x_{ij} = F(EX')F'$$

where e_{ij} is the indexes of employee i 's effectiveness in job j , E is the matrix of effectiveness, $F = [1 \ 1 \ 1 \ \cdots \ 1]$ (the number of 1 is n).

2.2.2 The Constraint Condition

The constraints are as follows:

- Every employee must have a job in the business.

$$\sum_{j=1}^n x_{ij} = 1$$

- Every job must be taken up by an employee.

$$\sum_{i=1}^n x_{ij} = 1$$

- The decision variable is a 0-1 variable, which means

$$x_{ij} \in \{0,1\}$$

2.2.3 The Primary Model

Considering the factors mentioned above, our model is as follows:

$$\text{max} \sum_{i=1}^n \sum_{j=1}^n x_{ij} e_{ij}$$

$$\begin{cases} \sum_{j=1}^n x_{ij} = 1 \\ \sum_{i=1}^n x_{ij} = 1 \\ x_{ij} \in \{0,1\} \end{cases}$$

2.3 The Intermediate Model

In practice, the actual situations we meet with are not always similar with the cases in **part 2.2**. In most cases, the supply of employees is greater than demand. Assume that the number of jobs is n , the number of job applicants is m and $m > n$. This means that there will be $(m - n)$ applicants who cannot get a job.

$$E = \begin{bmatrix} e_{11} & \cdots & e_{1n} \\ \vdots & \ddots & \vdots \\ e_{m1} & \cdots & e_{mn} \end{bmatrix}_{m \times n}$$

$$X = \begin{bmatrix} x_{11} & \cdots & x_{1n} \\ \vdots & \ddots & \vdots \\ x_{m1} & \cdots & x_{mn} \end{bmatrix}_{m \times n}$$

In this model, we only need to consider the constraint that every job is taken up by employees.

$$\max \sum_{i=1}^n \sum_{j=1}^n x_{ij} e_{ij} = F(EX')F'$$

$$\begin{cases} \sum_{i=1}^n x_{ij} = 1 \\ \sum_{j=1}^n x_{ij} \leq 1 \\ x_{ij} \in \{0,1\} \end{cases}$$

where F is an m -order row vector.

2.4 The Advanced Model

In actual situations, we often have to consider some contingency factors. These

contingency factors will limit the management and allocation of human resources. For instance, some employees may not have the ability to do certain jobs or some can do only a part of the jobs. We need to add the following constraints into the model:

$$x_{ij} = d$$

where d is a 0-1 variable.

$$d = \begin{cases} 0 & \text{employee } i \text{ does not take up job } j \\ 1 & \text{employee } i \text{ takes up job } j \end{cases}$$

Similarly, there may be factors of inter-relationship between employees. For example, some employees may finish their work only if they work together with certain other employee or they may only do their work if they are separated from certain other employees. Considering this, we add the following constraints:

$$f(X) = 0$$

where $f(X)$ is the equation of limitations of inter-relationship of matrix X .

3 Model II: The Shortest Queue of Job Applicants

The hermit crabs' behavior gives us much enlightenment. When they find a beautiful shell, they will firstly wait for a certain period of time and make sure that no other hermit crabs suit the shell more than themselves. Then they will claim the shell as their own shell. In business human resources management, this process is even more complex. After we make sure that the structure of human resources within the business achieves optimization, we will consider recruiting new employees from the society. We cannot learn about the information about all the applicants in the society before recruitment since this will bring huge costs to the business. Similar to hermit crabs' behavior that they will wait for only a certain period of time, we must determine a proper length of the queue of job applicants. This length is the number of job applicants about whom we need learn information in detail. And we should assume that when we set this length, we know nothing about all the job applicants.

Based on the analysis above, we assume that the condition of recruitment is as

follows:

- The total number of job applicants in the society is much larger than the number in the queue.
- The ability structure of applicants in the society obeys normal distribution.
- The total time required in recruiting employees is fixed. And the expenditures of recruitment is related to the length of the applicant queue.

3.1 The Distribution of Human Resources

We assume that the ability of job applicants obey normal distribution

$$b \sim N(0, \sigma^2)$$

where b is the ability of each applicant.

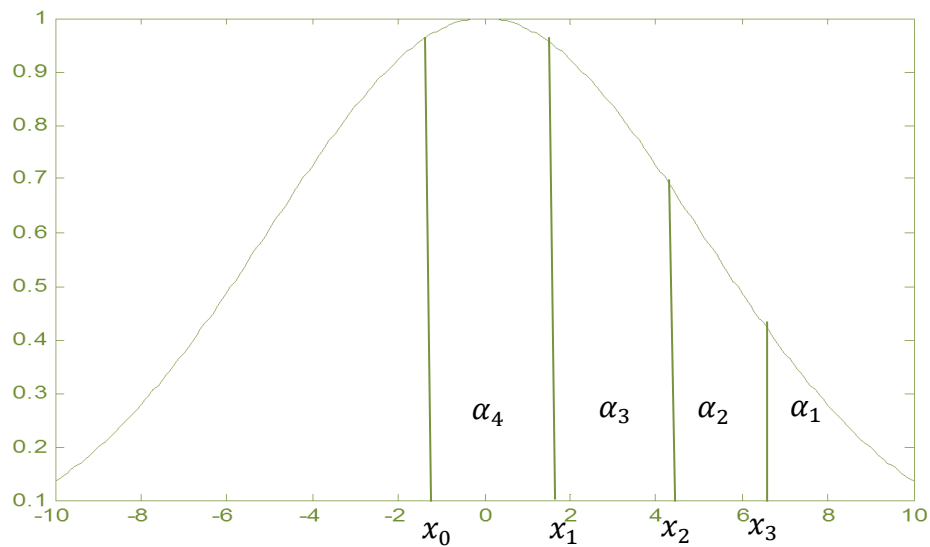


Figure 2: Distribution of HR

As is shown in Figure 2, x_0 is the lower limit of job ability required for the applicants. Here, x_0, x_1, x_2, x_3 form four ranges, dividing the ability of the applicants into four degrees. In the diagram of the normal distribution, section A, B, C and D represent the degrees of working ability in Model I, which is $\{2,3,4,5\}$, shown

as follows in Table 2:

Table 2: Range of Effectiveness Level

<i>Range</i>	<i>Degree of working ability</i>	<i>Probability</i>
$[x_0, x_1)$	D	α_4
$[x_1, x_2)$	C	α_3
$[x_2, x_3)$	B	α_2
$[x_3, x_4)$	A	α_1

3.2 Effectiveness-cost Model

3.2.1 Total Effectiveness

According to the conditions above, we define that the length of the applicant queue is L . Under the condition that we only know about the information of the first L applicants, our goal is to select the best one applicant as our employee from the applicants in the queue. We set up a matrix of random probability as follows

$$Prob = [Prob_1 \quad Prob_2 \quad Prob_3 \quad Prob_4]$$

where $Prob_i$ is the probability that the best applicant in the queue with a length of L falls in the degree of i .

$$\begin{cases} Prob_1 = 1 - (1 - \alpha_1)^L \\ Prob_2 = 1 - (1 - \alpha_2)^L \\ Prob_3 = 1 - (1 - \alpha_3)^L \\ Prob_4 = 1 - (1 - \alpha_4)^L \end{cases}$$

Then we define that Y is the total effectiveness.

$$E(Y) = [Prob_1 \quad Prob_2 \quad Prob_3 \quad Prob_4] \begin{bmatrix} e_1 \\ e_2 \\ e_3 \\ e_4 \end{bmatrix}$$

where $E(Y)$ is the expectation of total effectiveness, e_i is the index of working effectiveness of employees in the degree of i .

$$E(Y) = \sum_{i=1}^4 (1 - (1 - \alpha_i)^L) e_i$$

3.2.2 The Cost Curve

In this model, we consider that the extension of the applicant queue will increase expenditures. These include the expenditures on material resources and human resources required in learning about the information about applicants. The expenditures are in direct proportion to the length of the applicant queue.

$$E(C) = \delta L + \beta$$

where C is the cost, δ is a parameter.

δ can be estimated by the historical cost happened in the past in the process of recruitment within the business.

Then we define that W is value measured by money generated by employees under the unit index of effectiveness in unit time.

$$E(V) = \frac{1}{W} E(C) = \frac{1}{W} (\delta L + \beta)$$

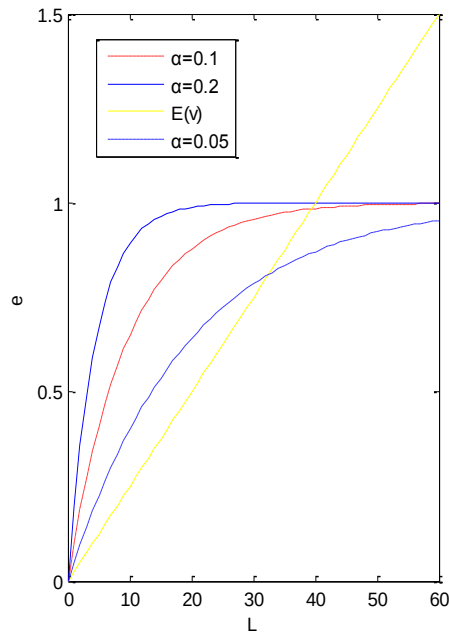
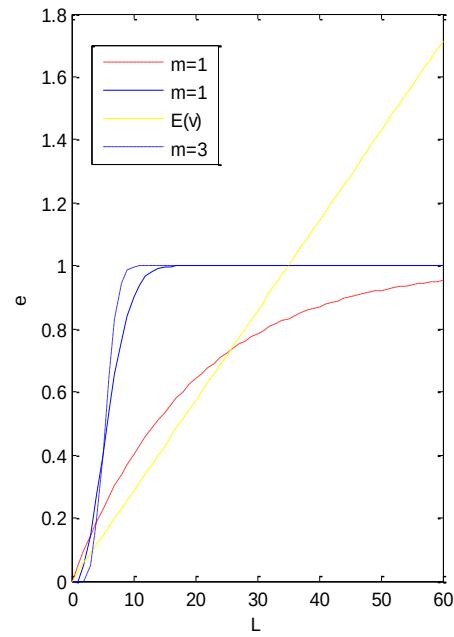
where V is the reimbursed effectiveness generated from the process of recruitment.

3.2.3 The Optimal Length of Applicant Queue

When the marginal effectiveness is equal to marginal cost, we can get the optimal length of applicant queue.

$$\frac{dE(Y)}{dL} = \frac{dE(V)}{dL}$$

This can be shown in Figure 3 and Figure 4:

Figure 3: Influence of α_i Figure 4: Influence of m

3.2.4 The Model of Selections of more than one applicants

The analysis above is mainly used to solve the problem when the enterprise needs to recruit only one new employee. When the number of new employees they need to hire is m , we can set up the following matrix of probability:

$$Prob = \begin{bmatrix} Prob_{11} & Prob_{12} & Prob_{13} & Prob_{14} \\ \vdots & \vdots & \vdots & \vdots \\ Prob_{m1} & Prob_{m2} & Prob_{m3} & Prob_{m4} \end{bmatrix}_{m \times 4}$$

where

$$\begin{cases} Prob_{i1} = (1 - (1 - \alpha_{i1})^{c_L^m}) \\ Prob_{i2} = (1 - (1 - \alpha_{i2})^{c_L^m}) \\ Prob_{i3} = (1 - (1 - \alpha_{i3})^{c_L^m}) \\ Prob_{i4} = (1 - (1 - \alpha_{i4})^{c_L^m}) \end{cases}$$

The expectation of total effectiveness will be as follows:

$$E(Y) = \sum_{i=1}^m \sum_{j=1}^4 Prob_{ij} e_j$$

Here the cost curve is the same with that in the situation where only one new

employee is recruited.

4 Model III: The Dynamic Model

In model I, we analyzed the allocation of human resources under the condition that the number of employees and jobs are the same and fixed. In model II, we analyzed the choice of the best applicants and the reduction of expenditures in the process of recruitment. However, in practice, the allocation of human resources is liquid and dynamic. In order to simulate this dynamic process, we must know the links, steps and directions of human resources allocation. Then we can apply the method of vacancy chain from hermit crabs to the model of dynamic human resources optimization and management.

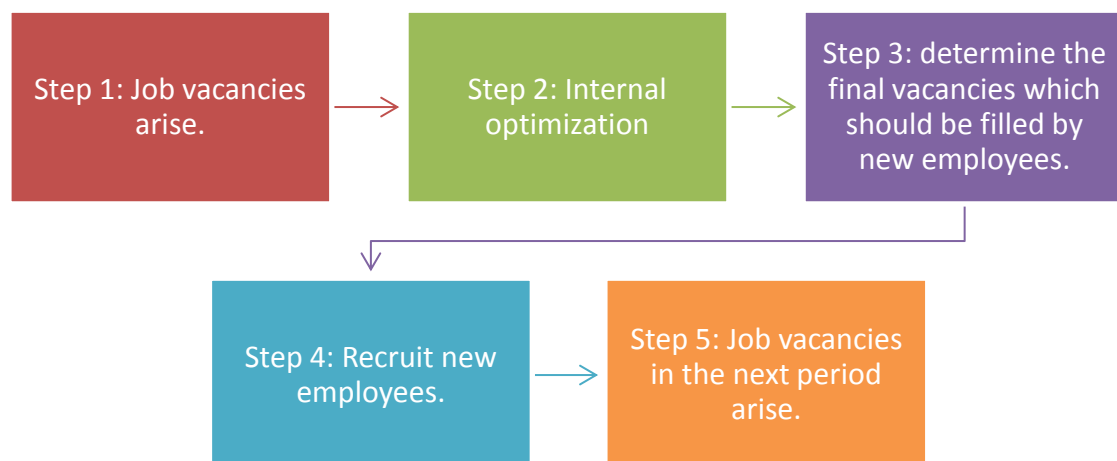


Figure 5: Steps of Model III

In this dynamic process, initial job vacancies are the driving force of adjustment of jobs within the business. Then the optimization will be carried out through the method in model I. After a series of job exchanges, the job vacancies left will be filled by newly recruited employees from the society. After this, in the next period when existing employees leave their jobs and new vacancies arise, the adjustment will be repeated.

Our goal is to make sure that in the dynamic process of job exchange and recruitment, higher effectiveness and lower rate of staff turnover can be achieved at the same time.

We can show the process of job adjustment using the following diagram:

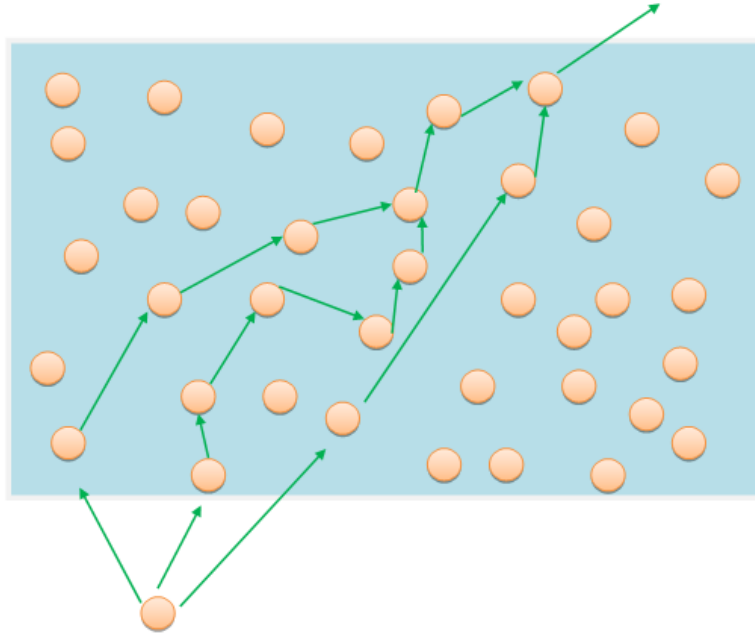


Figure 6 : Chain of job exchanging

4.1 The Times of Job Exchange

Let's assume that in an enterprise the total number of jobs is constant. Before the job reallocation, the optimal job allocation matrix is as follows:

$$X(t) = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1m} \\ x_{21} & x_{22} & \cdots & x_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{nn} \end{bmatrix}$$

The matrix of effectiveness is as follows:

$$E(t) = \begin{bmatrix} e_{11} & e_{12} & \cdots & e_{1m} \\ e_{21} & e_{22} & \cdots & e_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ e_{n1} & e_{n2} & \cdots & e_{nn} \end{bmatrix}$$

Now there is the phenomenon that some existing employees will leave the enterprise. Assume that the number of employees leaving their jobs and newly recruited employees is q . Replace the indexes of effectiveness of the employees who left the enterprise with that of the new employees. Then there be $E(t + 1)$ and we can define

new optimal job allocation matrix $X(t + 1)$.

Apply these factors into model I and we can get:

$$\begin{aligned} \max \sum_{i=1}^n \sum_{j=1}^n x(t+1)_{ij} e(t+1)_{ij} \\ \begin{cases} \sum_{j=1}^n x(t+1)_{ij} = 1 \\ \sum_{i=1}^n x(t+1)_{ij} = 1 \\ x(t+1)_{ij} \in \{0,1\} \end{cases} \end{aligned}$$

Then in period $t + 1$, the optimal job allocation and the times of transfer of jobs will be

$$k = \sum_{i=1}^n \sum_{j=1}^n \frac{|x(t+1)_{ij} - x(t)_{ij}|}{2}$$

In the equation above, due to the particularity of matrix $E(t + 1)$ and $E(t)$, we can solve the times of transfer of jobs by mathematical transformations.

4.2 The Shortest Transfer Path of Jobs

In the practical process of human resources management, due to the cost of transfers of jobs within the group, we will generally try to reduce the times and range of transfers and keep the effectiveness at a high level. Therefore, the objective function in period $t + 1$ can be as follows:

$$\begin{cases} \min k \\ \max \sum_{i=1}^n \sum_{j=1}^n x(t+1)_{ij} e(t+1)_{ij} \end{cases}$$

The constraints are

$$\begin{cases} \sum_{j=1}^n x(t+1)_{ij} = 1 \\ \sum_{i=1}^n x(t+1)_{ij} = 1 \\ x(t+1)_{ij} \in \{0,1\} \end{cases}$$

With the help of software Lingo, we can solve $X(t+1)$, which will also be used as the conditions for $X(t+2)$. This will guarantee that in each adjustment of human resources, our goal is to achieve the shortest transfer path and the highest total effectiveness.

4.3 The Prediction of Transfer Effectiveness

In practical situations, the appearance of initial job vacancies is random. The indexes of newly recruited employees' effectiveness are random, too. Therefore, the enterprise must predict the trend of human resources in the society for the next period.

4.3.1 Transfer of a Single Employee

In order to simplify the model, we will firstly look at the prediction in which only one employee is changed for one time.

We make the following assumptions

- The total number of existing employees is n , of which the number of employees required to be hired from the society is m .
- $A(t)$ is the effectiveness matrix of existing employees. $A'(t)$ is the effectiveness matrix of applicants. $X(t)$ is the allocation of existing employees.
- The probability of leaving the jobs is the same for all the existing employees. The probability of recruitment is the same for all the applicants.

Then the probability of leaving the jobs for existing employees is

$$Prob_1 = \frac{1}{n}$$

The probability of recruitment of applicants is

$$Prob_2 = \frac{1}{m}$$

Then we can get the following matrixes:

$$Y = \begin{bmatrix} y_{11} & y_{12} & \cdots & y_{1n} \\ y_{21} & y_{22} & \cdots & y_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ y_{m1} & y_{m2} & \cdots & y_{mn} \end{bmatrix}$$

$$K = \begin{bmatrix} k_{11} & k_{12} & \cdots & k_{1n} \\ k_{21} & k_{22} & \cdots & k_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ k_{m1} & k_{m2} & \cdots & k_{mn} \end{bmatrix}$$

where y_{ij} is the total effectiveness generated when applicant i takes the place of existing employee j . We can use model 4.2 to solve each y_{ij} in matrix Y . k_{ij} is least times of job transfers when applicant i takes the place of existing employee j .

Therefore, we can get

$$E(Y) = \frac{1}{mn} F Y F'$$

$$E(K) = \frac{1}{mn} F K F'$$

4.3.2 Transfer of more than one Employees

If more than one employees need to be recruited, the number of probable transfers will be $S = C_m^g C_n^g$, where g is the number of employees need to be recruited from the society.

Then we can get the following equations:

$$E(Y) = \sum_{i=1}^S y_i Prob_i$$

$$E(K) = \sum_{i=1}^S k_i Prob_i$$

4.3.3 Uncertain Transfer of Employees

Under the condition of transfers of more than one employees, if $m < n$, the number of probable transfers will be

$$C_m^1 C_n^1 + C_m^2 C_n^2 + \dots + C_m^m C_n^m$$

Let $S = C_m^1 + C_m^2 + \dots + C_m^m$. Then Y and K will be S -order vectors. And expression of $E(Y)$ and $E(K)$ will be the same as the equations in 4.3.2

4.4 Predictions of the Change of HR Expenditures

4.4.1 The goal of Predictions

To predict the situation of human resources in period t , we are mainly going to know about the job transfer path and total effectiveness in period t . Through the method in 4.3, we can get the expectation of times of transfer and total effectiveness in period $t + 1$ based on the job allocation matrix and effectiveness matrix in the basic period.

We define the following equation:

$$r = \frac{\sum_{i=1}^n \sum_{j=1}^n e_{ij} x(t+1)_{ij}}{k}$$

where r is the ratio between total effectiveness and times of transfer. We consider that the result and efficiency of human resources allocation will be better if the value of r is larger.

4.4.2 The Prediction of Transfer Path

The expectation of efficiency of human resources allocation in period $t + 1$ is:

$$E(r) = \frac{E(y)}{E(k)}$$

In terms of the prediction of transfer path, we refer to the grey prediction model. We use the combination which is the closest to the efficiency of human resources

allocation and $E(r)$ as follows:

$$\min\{|E(r) - r(1)|, |E(r) - r(2)|, \dots, |E(r) - r(q)|\}$$

We should find the one which is the closest to the expectation as the initial human resources allocation in period $t + 1$.

Then based on the analysis in period $t + 1$, we can continue to predict the management of human resources in period $t + 2$ and so on. Our prediction of transfer path is led by $E(r)$.

5 Analysis of Solutions and Result

In order to verify the availability of the model precisely, we use computers to simulate the dynamic process of human resources management.

Firstly, we assume that there are differences between working effectiveness of different employees. Now we define two types of human resources allocation:

a) $\alpha_1 : \alpha_2 : \alpha_3 : \alpha_4 = 0.2 : 0.3 : 0.4 : 0.1$

b) $\alpha_1 : \alpha_2 : \alpha_3 : \alpha_4 = 0.1 : 0.2 : 0.4 : 0.3$

It is obvious that human resources allocation type a) is better than type b).

For some enterprises, the effectiveness of existing employees is worse than that of the newly recruited employees from the society. For some other enterprises, the average level of working ability of newly recruited employees is higher than that of existing employees.

With the help of computers, we simulate the total effectiveness human resources and times of job transfers in the next 5 years under the condition of two types of human resources allocation respectively.

In the process of simulation, we establish the random effectiveness matrix of existing employees and newly recruited employees according to the weight mentioned above

respectively. Then we can use the method in model III to analyze and predict the change of human resources in the future.

We set up four kinds of random patterns to simulate the practical human resources adjustment.

- Pattern 1: The period of adjustment is 1 year. The number of employees changed is 1. The average effectiveness level of existing employees is lower than that of newly recruited employees.
- Pattern 2: The period of adjustment is 1 year. The number of employees changed is 2. The average effectiveness level of existing employees is lower than that of newly recruited employees.
- Pattern 3: The period of adjustment is 1 year. The number of employees changed is 1. The average effectiveness level of existing employees is higher than that of newly recruited employees.
- Pattern 4: The period of adjustment is 1 year. The number of employees changed is 2. The average effectiveness level of existing employees is higher than that of newly recruited employees.

Table 4: expectations of Different Patterns

	Pattern 1		Pattern 2		Pattern 3		Pattern 4	
	E(y)	E(k)	E(y)	E(k)	E(y)	E(k)	E(y)	E(k)
Year 1	0.7495	3	0.7459	4	0.8762	3	0.8762	4
Year 2	0.7958	2.7	0.8265	3.3	0.8233	2.7	0.8012	3.2
Year 3	0.8433	2.1	0.8587	2.8	0.7903	2.1	0.7732	2.7
Year 4	0.8608	1.6	0.8658	2.5	0.7801	1.6	0.7603	2.3
Year 5	0.8658	1.4	0.8708	2.4	0.7765	1.4	0.7545	2.2
Long-term trend	0.9008	1	0.9008	2	0.7459	1	0.7459	2

5.1 Analysis of the result

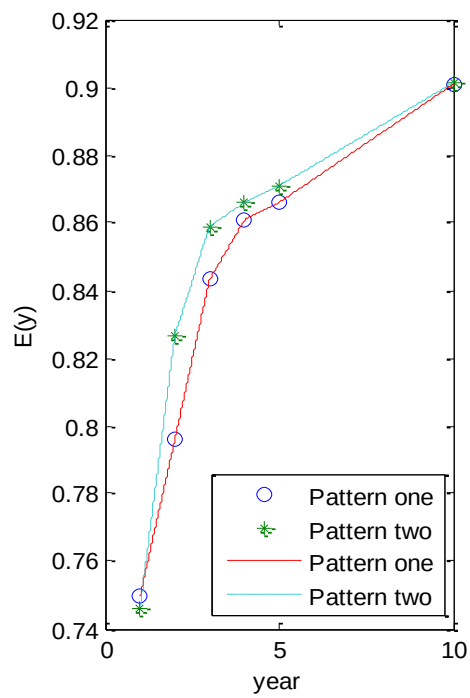


Figure 7.1

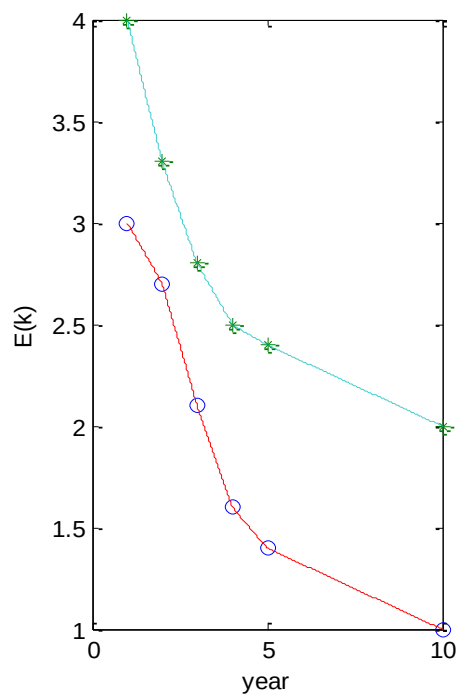


Figure 7.2

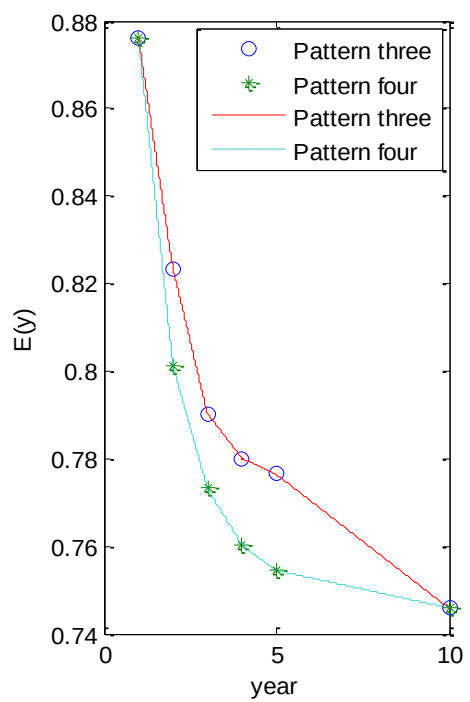


Figure 7.3

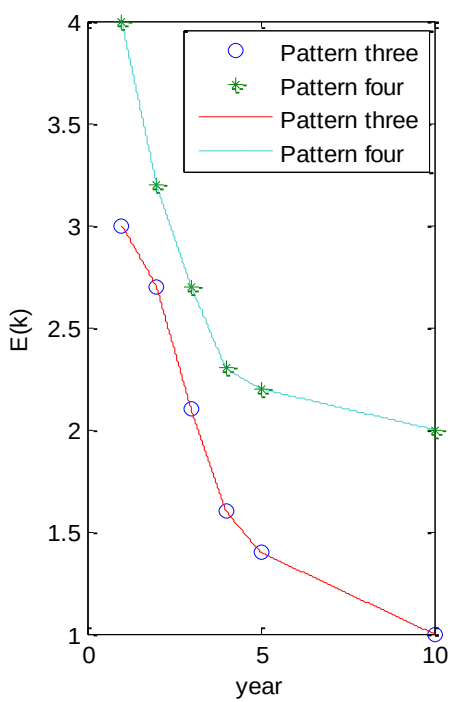


Figure 7.4

Figure 7: changes of Expectations with Time

From Figure 7.1 and 7.2, we can know that when the newly recruited employees' ability is better than the existing employees, the total effectiveness of the enterprise as whole will show a upward trend as time goes by. But the speed of the increase will slow down and tend to be close to the expectation of the optimal effectiveness of the applicants in the society.

In the meantime, with the process of the combination of existing employees and new employees, the expectation of transfer times in each period tends to be close to 1. To put it another way, internal adjustment is not required since there is no distinction between the effectiveness of existing employees and new employee. Therefore there is no need to transfer jobs within the business.

Situations are different in pattern 3 and 4. In these two patterns, outside applicants' effectiveness is lower than existing employees. In fact job vacancies in these situations are the loss of human resources. Therefore, total effectiveness of optimal allocation will show a downward trend until the total effectiveness of existing employees is the same with that of those in the society.

In all the patterns, times of job transfers will decrease with the pass of time due to the reduced to differences between ability of existing employees and other applicants.

In each period, the number of transfers of employees directly determines the speed of human resources allocation. So it can be seen from graph 3 and graph 4 that the patterns in which 2 new employees are recruited show a faster speed of transfers than other patterns.

5.2 Solutions of the Problem

5.2.1 Answer to Question 1

Our model aims at seeking employees and applicants of high effectiveness. This is consistent with the fact in the process of recruitment. But if we want to apply this model to all industries, the following conditions should be met:

- The applicants for all kinds of jobs are sufficient.
- There is no limitation on transfers of employees within the enterprise.

5.2.2 Answer to Question 2

Considering all kinds of factors which will influence the length of the applicants queue, we use random arrays to verify the model and predict the change of total effectiveness and times of transfers in the following 5 years under different assumptions.

6 Evaluation of the Model and Possible Improvements

In our essay, we divide the models into 3 parts: the static human resources programming, the shortest queue of job applicants and the dynamic model of human resources management. The combination of these models provide the enterprise with available strategies to pursue the goal of optimal human resources management. In addition, the result and solutions from the model are reliable and consistent with the basic theory of business staff management. Therefore, the feasibility of these models are high in practice.

6.1 Advantages of Our Models

- Simplify complex practical problems into mathematical problems. Divide the process of human resources allocation into 3 stages and use the partial optimization to deduce the optimization of the group as a whole.
- The assumptions of the dynamic model are reasonable. For instance, job vacancies will firstly be filled by existing employees and then new employees.
- The objective of the model is to maximize the total effectiveness of the enterprise as a whole.
- The abstract process of evaluation of performance at work is transferred to a mathematical problem.

- In the models, plenty of random probability methods are used. This makes the models more applicable to real life.

6.2 Disadvantages of Our Models

- Our models are based on the assumption that employers know all the information about employees and applicants. But in practice, our information is usually not so comprehensive. To avoid information asymmetry is one of our goals in the future.
- In the models we assume that each employee's effectiveness index remains unchanged. This is reasonable in a short period. However, with the learning curve effect and the influence of competition, everyone's working effectiveness is increasing. And this change varies in different individuals. In addition, some employees' working effectiveness will decrease with the increase of their age.

6.3 Possible Improvements

6.3.1 Application of Random Arrays

In order to reduce the effect of incorrect information and save expenditures, we simplify the measure of individual effectiveness index as the structure of effectiveness of the whole group. Then different matrixes of random arrays can be generated according to their distributions. We have used this method in the process of simulation which is of great efficiency.

6.3.2 The Change of effectiveness index

In the long term, each employee's effectiveness index towards every job is a variable. It is mainly influenced by the following factors:

- Learning curve effect
- Level of experience
- Physical ability and the decrease of ability with the increase of ages

- Personal motives and other negative moods

The first two factors can increase the effectiveness index while the last two will limit it. Their total impact on the effectiveness index is shown by the factor of time.

We define:

$$e(t)_{ij} = f(t)$$

The forms of $f(t)$ will differ in different jobs. Shown in the following graphs are certain types of effectiveness curves:

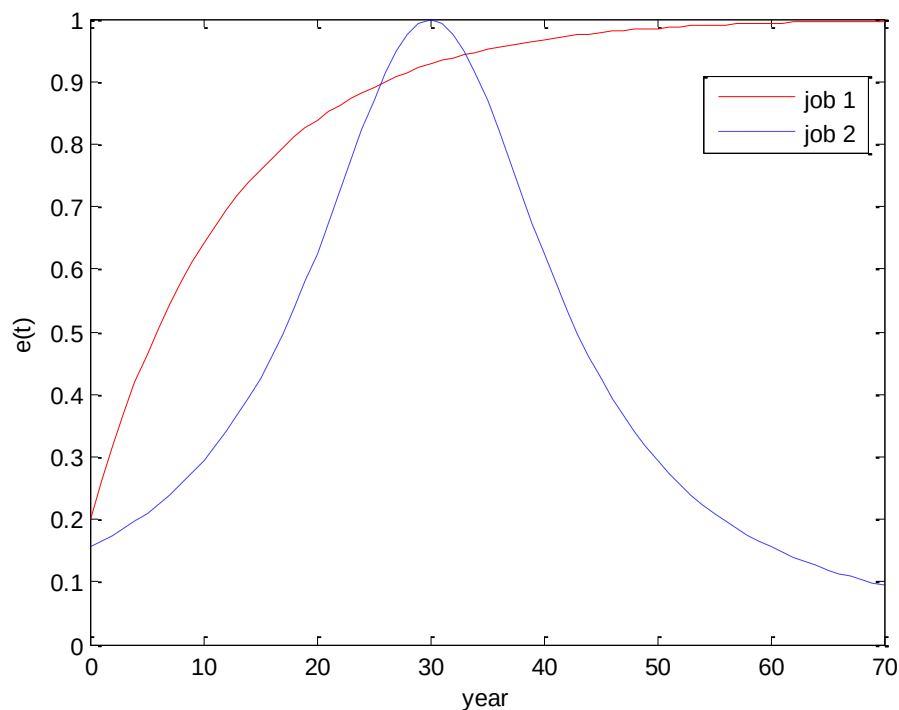


Figure 8: Effectiveness Curve

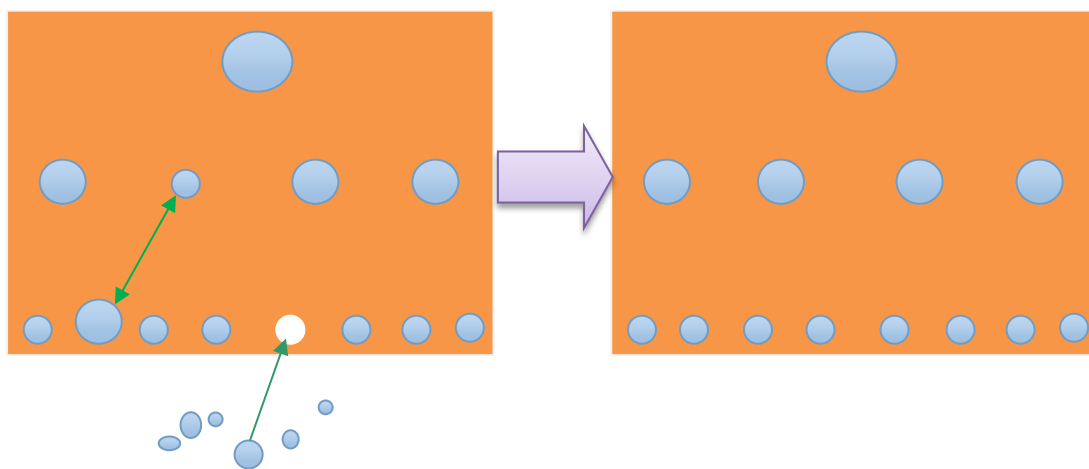
If we consider the dynamic change of effectiveness index, the models will be improved a lot and the errors will be reduced as well.

7 Advertising Sheet

Best Friends of HR

Are you still confused about recruitment? Are you still struggling to work out the number of applicants required? Come here and visit our Human Resource website.

Here, easy realization of optimal human resources management will never be a distant dream. Feel attracted? Do not hesitate!



Reasons to choose us:

- Comprehensive factors and indexes are used to measure the effectiveness in job rotation and recruitment.
- The methods used are dynamic.
- Both the benefits of employers and applicants are considered.
- Take account of the internal and external factors of staff changes.

Our efforts:

- Deeply analyze all the situations which trouble the employers in the process of recruitment.
- Take full account of factors influencing employees' working effectiveness.
- Set up 3 models and simplify the process of recruitment

8 References

- [1] Persky, L., Felsenstein, D. Restricting access in a job chains model of local employment creation[J]. *Annual Regional Science*, 2006(40): 423-435
- [2] Zipkin, P. Mortgages and markov chain: a simplified valuation model[J]. *Management Science*, 1993, 39(6): 683—691
- [3] Stmyk, R. J. Understanding housing vacancy rate in a developing country: Jordan[J] *The Journal of Development Areas*, 1988(22): 373—380
- [4] Weinberg, D. H, Friedman J. & Mayo, S. K. Interurban residential mobility: the role of transaction costs, market imperfections and household disequilibrium[J]. *Journal of Urban Economics*, 1981(9): 332-348
- [5] Wheaton, W. C. Vacancy, search, and price in a housing market matching model [J] *Journal of Political Economy*, 1990(98): 1270—1292
- [6] Rosenthal, L. Chain—formation in the owner-occupied housing market[J]. *The Economic Journal*, 1997(107): 475—488
- [7] Persky, L., Felsenstein, D. & Carlson, V Job vacancy chain in metropolitan labor markets[J] *European Congress of the Regional Science Association*, 2002, 42(1): 27—31
- [8] Sands, G A model for the evaluation of filtering [J] *Growth and Change*, 1977(10): 20 — 24
- [9] Dzur, R. & Romsa, G. Housing construction vacancy chains and residential mobility in Windsor[J] *The Canadian Geographer*: 1977(21): 223-236
- [10] Winstanley, A., Thorns, D. C. & Perkins, H C. Moving house, creating home: exploring residential mobility[J] *Housing Studies*: 2002, 17(6): 813-832
- [11] Schettkat R (1996) Flows in labor markets: concepts and international comparative results, 14–36. In Schettkat R (ed.) *The flow analysis of labor markets*.

Routledge, London

[12] Emmi, C p. & Magnusson, L. Issues in the use design of residential vacancy chain models[J]. The National Swedish Institute for Building Research, 1992

[13] Clark, W A. v, Markov chain analysis in Geography: an application to the movement of rental housing areas[J]. Annals of the Association of American Geographers, 1965(55): 351—359

[14] Bysveen, T. & Knutsen, S. Vacancy chains initiated by out-migration: a study of the housing market in Oslo[J]. Housing Studies, 1987(2): 203-212

[15] Sweeney, J. L. Quality, commodity hierarchies, and housing markets[J].

Econometrics, 1974(42): 147-167

[16] Tumer, L. M. Who gets what and why? Vacancy chains in Stockholm's housing market[J]. European Journal of Housing Policy, 2008, 8(1): 1-19

9 Appendix

Programs and codes

2.0-2.5(Lingo)

```

model:
sets:
t/1..10/;
w/1..10/;
links(t,w):c,x1;
endsets
data:
!c=(n-order matrix);
enddata

[obj]max=@sum(links(i,j):c(i,j)*x1(i,j));
@for(t(i):@sum(w(j):x1(i,j))=1);
@for(w(j):@sum(t(i):x1(i,j))=1);
@for(links(i,j):@bin(x1(i,j));x1(i,j)>=0);
end

```

3.1-3.2(Matlab)

```

l=0:1:60;
f4=1-(1-0.1).^l;
f5=1-(1-0.2).^l;
f8=1/40*l;
f9=1/35*l;
f1=1-(1-0.05).^l;
f2=1-(1-0.05).^(l.*(1-1)./2);
f3=1-(1-0.05).^(l.*(1-1).*(1-2)./6);
subplot(1,2,1)
plot(l,f4,'r--',l,f5,'b-',l,f8,'y-',l,f1,'b--')
subplot(1,2,2)
plot(l,f1,'r--',l,f2,'b-',l,f9,'y-',l,f3,'b--')

```

4.1-4.3

1.random matrix

2.(lingo)

```

model:
sets:
t/1..10/;
w/1..10/;
links(t,w):c,x1,x2;
endsets

```

```

data:
!c=?(n-order matrix);
!x2=?( n-order matrix;
enddata
[obj]max=@sum(links(i,j):c(i,j)*x1(i,j))-0.1*(@sum(links(i,j):@abs(x1
(i,j)-x2(i,j))));
@for(t(i):@sum(w(j):x1(i,j))=1);
@for(w(j):@sum(t(i):x1(i,j))=1);
@for(links(i,j):@bin(x1(i,j));x1(i,j)>=0;);
End

```

5.0(Matlab)

```

t=[1 2 3 4 5 10];
y1=[0.8762 0.8233 0.7903 0.7801 0.7765 0.7459];
ti=0:0.01:10;
y1i=interp1(t,y1,ti);
y2=[0.8762 0.8012 0.7732 0.7603 0.7545 0.7459];
y2i=interp1(t,y2,ti);
subplot(1,2,1)
plot(t,y1,'o',t,y2,'*',ti,y1i,'-',ti,y2i,'--')
y3=[3 2.7 2.1 1.6 1.4 1];
y4=[4 3.2 2.7 2.3 2.2 2];
y3i=interp1(t,y3,ti);
y4i=interp1(t,y4,ti);
subplot(1,2,2)
plot(t,y3,'o',t,y4,'*',ti,y3i,'-',ti,y4i,'--')

```

6.0(Matlab)

```

>> t=0:1:70;
>> y=1./(3./800.*t.^2-9./40.*t+4);
>> y3=1-0.8*exp(-0.08.*t);
>> y1=1./(1.6.*(3./800.*t.^2-9./40.*t+4));
>> plot(t,y3,'r-',t,y1,'b--')

```